

Joshua H. Hartlep

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EDUCATION

Duke University

Durham, NC

B.S. Electrical and Computer Engineering, B.S. Computer Science, GPA: 4.0/4.0

May 2028

- Relevant Coursework: Advanced Software Design, Algorithms, Probability, Machine Learning, Comp. Architecture

PROFESSIONAL EXPERIENCE

Asana (Incoming)

May 2026 – Aug. 2026

Software Engineering Intern — AI Retrieval Team

New York, NY

- Focusing on building and scaling **distributed AI infrastructure** to support low-latency retrieval and search capabilities for high-concurrency production workloads

Bridgewater Associates

April 2026

(Invite Only) — AI Innovation Hackathon

Westport, CT

- Selected via a competitive interview process for an invitation-only immersion designed for top-tier engineers and researchers
- Collaborated with leaders to build **agentic workflows and predictive systems** using custom AI tooling
- Shipped a functional prototype addressing complex market puzzles, moving from conceptual logic to production-ready code

Worthington Enterprises

May 2025 – Aug. 2025

Software & Systems Engineering Intern

Chilton, WI

- Developed high-performance **backend services in C++ and Python** to ingest and process real-time sensor and PLC data, enabling system-wide visibility into industrial operations
- Architected **anomaly detection logic** and scalable data pipelines to identify equipment faults and inefficiencies, reducing system downtime by **50%** and driving **\$800K+** in annual cost savings
- Optimized data throughput and reduced response latency by refining processing workflows, improving memory management, and eliminating bottlenecks in real-time data streams

PROJECTS

DukeDine – Nutrition Tracking Platform | [Demo](#) *Node.js, React Native, Python, Supabase, PostgreSQL*

- Engineered an automated **Python-based ingestion pipeline** to scrape and normalize nutritional data for **1,000+ items** from NetNutrition, populating a structured **Supabase Postgres** schema
- Architected an offline-first mobile experience using React Native (Expo) and SQLite, implementing a **TanStack Query sync queue** to ensure 100% data persistence during intermittent campus Wi-Fi connectivity
- Developed a multi-platform architecture comprising a React admin dashboard for menu management and an Express API to serve high-performance nutritional analytics to student users

Shari POS (Restaurant Platform) | [GitHub](#) | [Demo](#) *FastAPI, React, TypeScript, Docker, AWS, SQL*

- Developed a **full-stack POS system** with a multi-tenant ready PostgreSQL schema, supporting real-time table tracking and complex AYCE (All-You-Can-Eat) pricing logic.
- Built "The Lens," a custom analytics engine utilizing **raw SQL** and Python statistics to detect **revenue anomalies** ($|z| > 2$) and provide **A/B cohort comparisons** for restaurant managers.
- Orchestrated a **CI/CD pipeline** using Docker Compose and GitHub Actions for automated deployment to **AWS EC2**, integrating **S3** for secure storage and moderation of menu imagery.

TECHNICAL SKILLS

Languages: C, C++, Python, Java, SQL, TypeScript, Verilog, Assembly

Systems & Infrastructure: Linux/Unix, Distributed Systems, Cloud (AWS), Performance, REST APIs

AI/ML & Tools: Machine Learning, PyTorch, Agentic Workflows, Git, PostgreSQL, Node.js, React

Core Concepts: Real-time Systems, Scalable Architecture, Data Processing, Concurrent Programming, Algorithms